Case Study 12: Attendance Recovery Computation

1. Problem Statement

Calculate current attendance percentages based on total conducted and attended classes, determine eligibility against a threshold, and calculate the minimum consecutive classes required to reach eligibility under future attendance scenarios.

2. Objectives

- Determine current attendance status using fundamental arithmetic operators.
- Evaluate eligibility against a predefined mandatory percentage threshold.
- Calculate the exact count of consecutive future classes needed to recover from an attendance shortage.

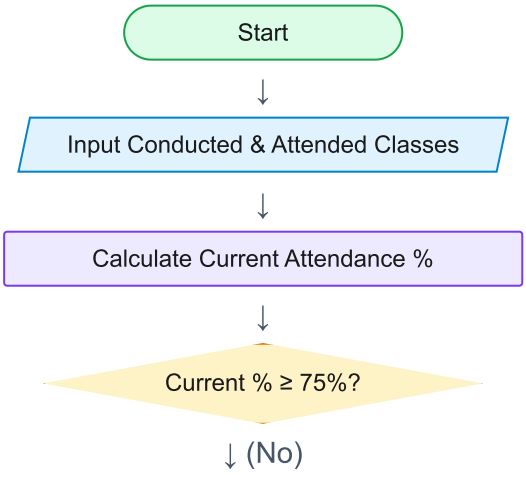
3. Assumptions

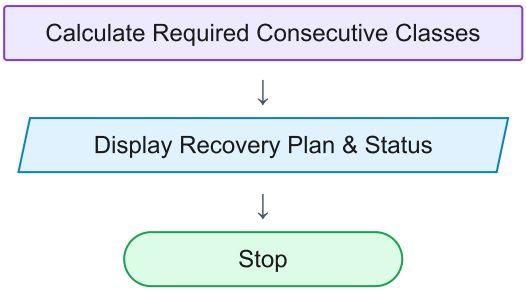
- Target eligibility threshold is set to **75%**.
- Total conducted classes must be greater than zero.
- Future recovery classes assume 100% attendance during that period.

4. Algorithm

1. Start.
2. Read `conducted` and `attended` class counts.
3. Calculate `current_pct = (attended / conducted) * 100`.
4. If `current_pct >= 75.0`, required extra classes = 0.
5. If `current_pct < 75.0`, calculate required consecutive classes `req` using formula:
   `req = ceil((0.75 * conducted - attended) / (1 - 0.75))`.
6. Display current attendance percentage, status, and recovery count.
7. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT conducted, attended target_pct = 75.0 IF conducted <= 0 OR attended < 0 OR attended > conducted THEN PRINT "Invalid input parameters" EXIT ENDIF current_pct = (attended / conducted) * 100.0 IF current_pct >= target_pct THEN status = "ELIGIBLE" req_classes = 0 ELSE status = "SHORTAGE" target_dec = target_pct / 100.0 num = (target_dec * conducted) - attended den = 1.0 - target_dec req_classes = CEIL(num / den) ENDIF PRINT current_pct, status, req_classes STOP
```

7. Complete Python Source Code

```
# Attendance Recovery Computation import math def calculate_recovery(conducted, attended, target_pct=75.0): if conducted <= 0 or attended < 0 or attended > conducted: print("Error: Invalid class counts provided.") return current_pct = (attended / conducted) * 100.0 target_dec = target_pct / 100.0 print("=" * 55) print(" ATTENDANCE RECOVERY REPORT") print("=" * 55) print(f"Conducted Classes : {conducted}") print(f"Attended Classes : {attended}") print(f"Current Percentage : {current_pct:.2f}%") print(f"Target Threshold : {target_pct:.1f}%") print("-" * 55) if current_pct >= target_pct: print("Status : ELIGIBLE") print("Recovery Plan : 0 extra classes needed.") else: num = (target_dec * conducted) - attended den = 1.0 - target_dec req_classes = math.ceil(num / den) print("Status : SHORTAGE") print(f"Recovery Plan : Must attend next {req_classes} consecutive classes.") new_attended = attended + req_classes new_conducted = conducted + req_classes new_pct = (new_attended / new_conducted) * 100.0 print(f"Projected Result : {new_attended}/{new_conducted} ({new_pct:.2f}%)") print("=" * 55) # Test Runs calculate_recovery(40, 24) print() calculate_recovery(50, 42)
```

8. Test Case and Expected Output

Test Case 1 (Shortage Case: Conducted = 40, Attended = 24):

```
     ========================================================= ATTENDANCE RECOVERY
REPORT ========================================================= Conducted Classes :
40 Attended Classes : 24 Current Percentage : 60.00% Target Threshold : 75.0% -----
------------------------------------------------ Status : SHORTAGE Recovery Plan
: Must attend next 24 consecutive classes. Projected Result : 48/64 (75.00%)
==========================================================
```

Test Case 2 (Eligible Case: Conducted = 50, Attended = 42):

```
     ========================================================= ATTENDANCE RECOVERY
REPORT ========================================================= Conducted Classes :
50 Attended Classes : 42 Current Percentage : 84.00% Target Threshold : 75.0% -----
------------------------------------------------ Status : ELIGIBLE Recovery Plan
: 0 extra classes needed. ============================================================
```

9. Conclusion

The Attendance Recovery Computation program provides a clear mathematical analysis of student attendance status, giving actionable insights into the exact number of consecutive classes required to satisfy eligibility criteria.